

## IPTV components

**Video encoders.** Video encoders transform an input stream that can be of various formats into a digital compressed stream. They are also responsible for encapsulating the video streams into a transport format.

**Video server.** Video servers are computer-based devices linked to large storage systems. Video content, previously encoded, is stored either on disk or in large banks of RAM. Video servers stream video and audio content through unicast or multicast to STBs.



IPTV components (Courtesy: IPTV India Forum)

**Middleware.** Middleware is the software and hardware infrastructure that conjoins the components of an IPTV solution. It is a distributed operating system that runs both on servers at the telco's location and on the STBs. It performs end-to-end configuration, provisions the video servers, links the electronic program guide (EPG) with the content, acts as a boot server for the STB and ensures that all STBs run compatible software. The key technical qualities of a middleware are reliability, scalability and the ability to interface with other systems.

**STB/terminal.** The STB is a CPE that facilitates interface with the user, the television and the network. For live TV and VoD, the STB supports an EPG that allows customers to navigate through the programming. The STB transforms a jumbled digital compressed signal into one that can be sent to the TV. The STB hosts the middleware and is the centre of the communications infrastructure within a home.

to real-time participation from viewers. Through this technology, viewers not only receive data but can also send data back. For example, e-mailing, instant messaging, ticket booking, e-shopping and participation in live votes are all possible through IPTV.

"With IPTV, you can order practically anything under the sun, from a pizza, to apparel, to tech gadgets, just by using your TV remote. It can enable polling on TV programmes like 'Indian Idol' as well; you don't need to write down lengthy phone numbers in order to send channels SMS messages or votes. All you have to do is simply press one of the buttons on the TV remote and it's done!" exclaims Sahu.

The number of applications that can be put on IPTV are virtually infinite, claim service providers. Service providers like USTarcom help in developing an ecosystem of such innovative application developers so that service providers can provide rich interactive services that have been tested to work well on the IPTV technology platform.

**Triple play service.** IPTV provides triple play service, which allows TV broadcasting, high-speed Internet connection and telephony on one platform.

**TV viewing at convenience.** IPTV gives viewers freedom from time-bound TV schedules. Viewers can now go back to watch any live broadcast they may have missed. Unlike DTH or cable TV, with IPTV, viewers don't need to give commands whenever they need to save their favourite programmes, since this burden now falls upon the service provider. Viewers also don't need to invest in expensive STBs with recording capabilities as the programmes are already saved on the servers of the service providers.

**On-demand content.** IPTV enables user-specific content delivery. A user can access content of any nature through this platform. For instance, IPTV offers VoD, which enables viewers to choose any video from a vast library and view it instantly. Similarly, viewers can listen to their favourite music, using the music-on-demand

service. Viewers can even pause, rewind and forward these videos at their convenience. Another example of on-demand content is education. IPTV has the capability to provide one-on-one tutoring to students through television.

**On-demand advertising.** This technology multiplies opportunities for advertisers as unlike other applications, wherein a viewer sees an advertisement and then goes to the store to find out more about the item or purchase it, through IPTV, a viewer has the facility to check out detailed information about the product instantly and, to a large extent, can also make the purchase there and then. This platform will therefore increase the rate of impulse buying. IPTV has the technology to track the real viewership of an advertisement.

Video calling, too, can be facilitated on an IPTV connection. Viewers can watch callers on their TV screens and also watch TV programmes at the same time.

**Internet penetration.** In a country like India, Internet usage is still very low. The fact that IPTV allows broadcast channels and Internet on one platform is a sure way of boosting Internet usage in the country.

## Hurdles to cross

Like any other new technology, IPTV too encounters certain challenges. For Sahu, the first challenge is the identification of the addressable market. "We believe IPTV patrons can be extended to two categories of people: those who already have a wireline or fixed line connection, which can be addressed easily, and those who need the connection. In a premise where a wire does not exist, it would be a challenge for us. The amount of copper lines in the country already amounts to 70 million, though voice users on wireline in the country are somewhere between 45-50 million. Once IPTV starts driving the growth of broadband, this extra unused wire will also become more valuable," says Sahu.

Others, however, find the high